

1. Module Name

03.0 Robot Framework

2. Experiment Title

Custom Keywords and Libraries in Robot Framework

3. Problem Statement

Create a Robot Framework automation suite using SeleniumLibrary and reusable custom keywords stored in a separate resource file. Automate opening SauceDemo, logging in, validating the Products page, logging out, and closing the browser.

4. Objective

To understand custom keywords, resource files, keyword reuse, SeleniumLibrary integration, and suite teardown in Robot Framework.

5. Tools, Software and Concepts Used

Python, Robot Framework, SeleniumLibrary, Selenium WebDriver, Chrome browser, custom keywords, resource files, suite teardown.

6. Source Code / Implementation Steps

1. Create a reusable resource file under resources/. 2. Import SeleniumLibrary. 3. Define custom keywords for opening the application, login, validation, logout, and cleanup. 4. Import the resource into the test suite. 5. Execute the test using `python -m robot custom_keywords.robot`.

```
*** Settings ***
Resource    resources/saucedemo_keywords.resource
Suite Teardown    Close SauceDemo Application

*** Test Cases ***
Custom Keywords Login Test
    Open SauceDemo Application
    Login To SauceDemo    standard_user    secret_sauce
    Verify Successful Login
    Logout From SauceDemo
```

7. Output / Labelled Screenshots

01_Robot_Custom_Keywords_Execution.png - Successful Robot Framework terminal execution showing the test result. The screenshot is stored in the Screenshots folder.

8. Result

The Robot Framework custom keyword test case executed successfully. The reusable keywords were invoked from the resource file and the SauceDemo login/logout flow was automated.

9. Observation

Resource files improve maintainability by keeping reusable Selenium actions separate from test cases. Custom keywords make the test case shorter and easier to read.

10. Conclusion

The experiment successfully demonstrated Robot Framework custom keywords, resource files, SeleniumLibrary integration, and reusable automation design.